

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. A software company discovers that its application crashes when users upload files larger than 100 MB. Developers modify the source code to eliminate the defect. Which type of software maintenance is performed?

- A. Preventive Maintenance
- B. Corrective Maintenance
- C. Perfective Maintenance
- D. Adaptive Maintenance

Answer: B

2. Which software characteristic explains why most software cost is associated with development rather than manufacturing?

- A. Software is physically assembled.
- B. Software engineering dominates its cost.
- C. Software requires expensive raw materials.
- D. Software production depends on factories.

Answer: B

3. Which SDLC model is most suitable when developers need frequent customer feedback because requirements are uncertain?

- A. Waterfall Model
- B. V-Model
- C. Prototype Model
- D. Linear Sequential Model

Answer: C

4. Which software process model delivers a working product through several successive releases?

- A. Waterfall Model
- B. Incremental Model
- C. V-Model
- D. Big Bang Model

Answer: B

5. Which statement is an example of a business requirement?

- A. The system shall encrypt passwords.
- B. The system shall respond within two seconds.
- C. Increase online sales by 25%.
- D. Users shall log in using email.

Answer: C

6. During Requirements Engineering, which phase confirms that documented requirements correctly represent stakeholder needs?

- A. Negotiation
- B. Validation
- C. Elaboration
- D. Elicitation

Answer: B

7. Which Requirements Engineering challenge usually results from vague statements that can be interpreted differently by different developers?

- A. Scope Creep
- B. Incomplete Requirements
- C. Ambiguous Requirements
- D. Requirement Baseline

Answer: C

8. Agile software development differs from traditional plan-driven development because Agile primarily focuses on:

- A. Comprehensive documentation
- B. Customer adaptability
- C. Fixed project scope
- D. Sequential execution

Answer: B

9. In Agile development, if additional work cannot fit within the current iteration, the recommended action is to:

- A. Extend the iteration.
- B. Increase team size.
- C. Reduce iteration scope.
- D. Cancel the Sprint.

Answer: C

10. Which XP value encourages developers to resist unrealistic project commitments?

- A. Respect
- B. Feedback
- C. Courage
- D. Simplicity

Answer: C

11. In XP, who determines whether a user story has been satisfactorily completed?

- A. Tester
- B. Customer
- C. Programmer
- D. Tracker

Answer: B

12. Which Scrum role is responsible for ensuring that Scrum practices and principles are correctly followed?

- A. Product Owner
- B. Customer
- C. Scrum Master
- D. Development Team Leader

Answer: C

13. During Scrum development, the Sprint Backlog is:

- A. Updated continuously by customers.
- B. Stable during the Sprint.
- C. Removed after Sprint Planning.
- D. Replaced daily.

Answer: B

14. Which DSDM principle recommends focusing on the most valuable 20% of requirements to obtain approximately 80% of business value?

- A. MoSCoW Rule
- B. 80:20 Rule
- C. Pair Programming
- D. Continuous Integration

Answer: B

15. In DSDM, requirements categorized as “Won’t Have” indicate features that:

- A. Must be implemented immediately.
- B. Are mandatory for project success.
- C. Are postponed for future releases.
- D. Have already been completed.

Answer: C

16. Which FDD process identifies business activities before creating individual software features?

- A. Build by Feature
- B. Design by Feature
- C. Build a Features List
- D. Code Inspection

Answer: C

17. According to FDD, which activity is mandatory before integrating completed features into the main project?

- A. Sprint Review
- B. Mandated Code Inspection
- C. Pair Programming
- D. Customer Demonstration

Answer: B

18. Which software myth incorrectly assumes that customers can provide only general objectives initially and complete requirements later without significant consequences?

- A. Management Myth
- B. Customer Myth
- C. Developer Myth
- D. Testing Myth

Answer: B

19. Which software failure is most likely when end users are not involved during software development?

- A. Improved maintainability
- B. Higher reliability
- C. User rejection of the system
- D. Faster deployment

Answer: C

20. Which practice aims to detect defects as early as possible by performing verification activities before implementation begins?

- A. Sprint Review
- B. Shift-Left Testing
- C. Pair Programming
- D. Refactoring

Answer: B